

CHANG LIU

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EDUCATION

University of Toronto <i>Incoming Ph.D. Student in Computer Science</i>	Toronto, Ontario	Starting 9/26
Johns Hopkins University (JHU) <i>M.S.E in Mechanical Engineering in Robotics, GPA: 3.74/4.00</i> Courses: Robot Devices, Kinematics, Dynamics and Control; Computer Integrated Surgery I/II; Magnetically Actuated and MRI-Compatible Robots; and Medical Robot System Design	Baltimore, Maryland	8/24-5/26
Chongqing University <i>B.E. in Mechanical Engineering and Automation, GPA: 88/100</i>	Chongqing, China	9/19-6/24
University of Cincinnati <i>B.S. in Mechanical Engineering, GPA: 3.82/4.00</i> Minor: Robotics and Automation Award: Dean's List (Fall 2020, Summer 2021, Spring 2022, Fall 2022, Fall 2023, and Spring 2024)	Cincinnati, Ohio	9/19-5/24

PROJECT EXPERIENCE AT JHU

Laboratory for Computational Sensing and Robotics (LCSR) <i>Digital Twin-Driven XR Teleoperation of dVRK using Meta Quest 3S (IROS 2026)</i> Advisor: Professor Peter Kazanzides	9/25-3/26
➤ Developed XR teleoperation loop in Unity, integrating dVRK XR visualization with Meta Quest 3S controller input for stable 6-DoF command execution and tactile button feedback. ➤ Built Python gateway (UDP) to stream real-time digital-twin object pose (JSON) into Unity, updating scene transforms with synchronized perception/control. ➤ Performed real-time object pose estimation using SAM3 for segmentation and FoundationPose for tracking	
Mobile Magnetic Manipulation Platform for Gastrointestinal Navigation with Deep Reinforcement Learning Controls (ISMR 2026) Advisor: Professor Axel Krieger	9/25-11/25
➤ Designed portable, low-cost magnetic manipulation platform using UR5 collaborative robot and compact four-electromagnet array for gastrointestinal navigation. ➤ Implemented Soft Actor-Critic (SAC) based sim-to-real Deep Reinforcement Learning (DRL) pipeline. ➤ Validated system by achieving millimeter-scale tracking accuracy.	
Time-Synchronized Multi-modal Data Collection Framework and Dataset for Surgical Robotics (ICRA 2026) Advisor: Professor Peter Kazanzides	2/25-9/25
➤ Built ROS/C++ online and offline recorders synchronizing stereo RGB/UV/depth, PSM/ECM kinematics, contact sensor topics, and side camera. ➤ Designed post-collection toolbox: stereo rectification, depth estimation, optical flow, and hand-eye-based kinematic reprojection to heatmaps. ➤ Demonstrated feasibility on skill assessment using multi-path regression framework with synchronized inputs	
Multi-Modal Data Collection for dVRK Instrument Segmentation Advisor: Professor Peter Kazanzides	2/25-9/25
➤ Integrated high-resolution Cornerstone stereo endoscope onto dVRK (da Vinci Research Kit), modifying dVRK kinematics and performed stereo calibration, stereo matching, and hand-eye calibration. ➤ Collected Multi-modal (RGB/UV/depth/kinematics) data and used Atracsys tracking system to obtain ground truth and to evaluate dVRK vs. hand-eye kinematic accuracy. ➤ Built UV-based labeling pipeline and deployed SAM-2 for instrument segmentation/keypoint tracking.	

PROJECT EXPERIENCE AT CHONGQING UNIVERSITY

State Key Laboratory of Mechanical Transmission <i>Polymer Gear System Conceptual Design and Optimization Research, Core Member</i>	9/22-9/23
➤ Developed multi-objective optimization models for polymer gear systems in turbofan engine transmissions. ➤ Contributed to contribution and correlation analysis code.	
"Gear life prediction model based on CTabGAN-XGBoost" Research, Group Leader	9/22-2/23
➤ Built data-augmentation pipeline (GAN+XGBoost) to predict polymer-gear fatigue life within triple error bands and applied SHAP interpretability to identify key factors influencing gear life.	

PAPERS AND PATENTS

- ***Chang Liu**, *Chenhao Yu, *Honghao Zhao, Hao Ding, Haochen Wei, Adnan Munawar, Mathias Unberath, and Peter Kazanzides. "Digital Twin-Driven VR Teleoperation with Multi-View Spatial Perception for Surgical Robots." **Accepted by** IEEE International Conference on Intelligent Robots and Systems (IROS), 2026. (***Equal contribution**)
- Open-H-Embodiment Consortium, including **Chang Liu**. "Open-H-Embodiment: A Large-Scale Dataset for Enabling Foundation Models in Medical Robotics." **Under review** at **Science Robotics**, 2026. [arXiv:2604.21017](https://arxiv.org/abs/2604.21017).
- *Zhifan Yan, ***Chang Liu**, *Yiyang Jiang, *Wenxuan Zheng, Xinhao Chen, and Axel Krieger. "A Mobile Magnetic Manipulation Platform for Gastrointestinal Navigation with Deep Reinforcement Learning Control." International Symposium on Medical Robotics (ISMR), 2026 (***Equal contribution**) DOI: [10.1109/ISMR69606.2026.11536616](https://doi.org/10.1109/ISMR69606.2026.11536616).
- *Haoying Zhou, ***Chang Liu**, Kevin Wu, Yimeng Wu, Zijian Wu, Yu Chung Lee, Sara Martuscelli, Septimiu E. Salcudean, Gregory S. Fischer, and Peter Kazanzides. "SurgSync: Time-Synchronized Multi-modal Data Collection Framework and Dataset for Surgical Robotics." IEEE International Conference on Robotics and Automation (ICRA), 2026. (***Equal contribution**) Website: [SurgSync: Time-Synchronized Multi-modal Data Collection](https://surgsync.github.io/)
- Zehua Lu, **Chang Liu**, Changjun Liao, Jiazan Zhu, Huaiju Liu, Yiming Chen, *Conceptual design and optimization of polymer gear system for low-thrust turbofan aero-engine accessory transmission*, **Journal of Computational Design and Engineering**, Volume 11, Issue 1, February 2024, pp.212-229, <https://doi.org/10.1093/jcde/qwae008>
- **Liu, C**, D. Wang, P. Chen, Y. Xie, & J. Dong. "Design a Wearable Exoskeleton to Free Litter Bearers' Hands and Achieve Multi-Task Capabilities for Casualty Evacuation." ASME International Mechanical Engineering Congress and Exposition (IMECE), 2024. <https://doi.org/10.1115/IMECE2024-145461>
- Chen, P, Y. Xie, **Liu, C**, D. Wang, & J. Dong. "Applying Binary Stiffness Mechanism to an Exoskeleton Arm for Casualty Evacuation." ASME International Mechanical Engineering Congress and Exposition (IMECE), 2024. <https://doi.org/10.1115/IMECE2024-145505>
- **Invention Patent**: A Predictive Model for Gear Fatigue Life and Its Establishment Method (CN119150470 A, publication date 17 Dec 2024; **Status**: under substantive examination)

PROFESSIONAL EXPERIENCE

Course Tutor (Paid)	9/25-12/25
Computer Integrated Surgery I (Taught by Professor Russell H. Taylor, LCSR, JHU)	
➤ Tutored on content covering calibration, registration, path planning, and robot kinematics.	
Graduate Research Assistant (Paid)	5/25-9/25
Professor Michael Kessler's Research Group (Materials Science & Engineering, JHU)	
➤ Modeled earplug-ear canal acoustics using COMSOL multiphysics (solid-acoustic coupling) to predict insertion loss under different material/geometry conditions.	
R&D Intern at Ethicon R&D Department	5/22-9/22
Johnson & Johnson Medical (Shanghai), Ltd.	Shanghai, China
	9/21-12/21
➤ Conducted suturing and leakage validation on porcine intestines and bronchi using multiple endoscopic stapler prototypes.	
➤ Designed endoscopic stapler test fixture and analyzed its feasibility with FEA finite element analysis.	
➤ Designed surgical suture apparatus, completed 2D and 3D drawings and 3D printing.	
ASME 2024 International Mechanical Engineering Congress and Exposition (IMECE 2024)	
➤ Oral presenter of two exoskeleton papers in IMECE 2024, Design and Control of Robots, Mechanisms & Structures sessions I & II	

COMPETITION

National 3rd Prize at China Internet+ College Student Innovation & Entrepreneurship Competition	8/22
Mathematical Contest in Modeling (MCM) & the Interdisciplinary Contest in Modeling (ICM) in the U.S.	
➤ Won the "H" award (the third prize at the international level)	1/22
National Innovation & Entrepreneurship Project (2022) – Core Member	
➤ Design of Adsorbable Aircraft in Complex Environment	4/22-4/23

TECHNICAL PROFICIENCY

Software and Robotics:	ROS1 and ROS2, Arduino, ANSYS, COMSOL, 3D Slicer, OpenCV, SolidWorks, and 3D printing
Programming Languages:	MATLAB, Python, and C++
Language Skills:	Chinese (Native) and English (Professional)